

Study Plan for Embedded Systems, PCB Design & Firmware Programming

Personal Study Plan

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Weekly Structure

Monday:

- 3-4 hours - Core Theory (Circuit Design, Microcontroller Concepts)
- Use: CircuitLab, PDFs, YouTube Lectures, Books

Tuesday:

- 2-3 hours - Simulations & Code Practice
- Use: Arduino IDE, ESP32, Logic Simulators

Wednesday:

- 4-5 hours - Deep Work Project Day (PCB Design, Schematic Capture, Firmware Projects)
- Use: KiCad, EasyEDA, GitHub

Thurs-Sun (Work Days):

- 20-30 min micro sessions (during commute or lunch)
- Flashcards (Anki), Podcast (Embedded.fm, The Amp Hour), YouTube (audio-only)

Monthly Milestones

Study Plan for Embedded Systems, PCB Design & Firmware Programming

Month 1: Circuit Theory & Simulation

- LED blink, voltage dividers (CircuitLab, Falstad)

Month 2: Microcontrollers & C

- Arduino Nano input/output, button debounce (Arduino IDE)

Month 3: PCB Design

- Schematic + Layout, power routing, DRC check (KiCad/EasyEDA)

Month 4: Embedded Projects

- ESP32 + OLED + sensor readout (PlatformIO)

Month 5: Firmware Focus

- I2C, UART, interrupts, timers

Month 6: PCBA

- Order board, solder, flash firmware, test

Environment & Tools

Desktop Setup:

- Dual Monitor (Design + Docs)
- KiCad, EasyEDA, CircuitLab, Falstad, Arduino IDE, PlatformIO

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Study Tools:

- Flashcards: Anki (spaced repetition)
- Tracker: Trello Board

Weekly Reflection Log (Template)

Date | What I Learned | What I Built | Issues Faced | Plan for Monday

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Repeat weekly for progress tracking.

Certifications to Pursue

- IPC-A-610 (Soldering Standards)
- IPC-7711/21 (Rework/Repair)
- Arduino Certification
- Embedded Systems (Coursera / edX)
- KiCad PCB Design (Udemy)
- ESP32 / IoT Specialization (Random Nerd Tutorials, UCSD)

Degree Programs to Consider

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- A.S. Electronics Engineering Technology (local or online)
- B.S. Embedded Systems / Electrical Engineering (WGU, ASU Online, Purdue Global)
- MITx or edX MicroMasters in Embedded Systems

Quote

"Design to Understand, Build to Remember."